

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT REPORT

ITA04 — STATISTICS WITH R PROGRAMMING

Customer Churn Analytics and Predictive Modelling Using R

ShopSmart Retail Case Study

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1. Problem Statement

ShopSmart Retail is a mid-sized e-commerce company that wants to reduce customer churn, defined in this assignment as customers who stop purchasing. The available raw export is transaction-level data containing customer ID, purchase date, product category and subcategory, amount spent, and a churn indicator.

The engineering and analytics challenge is to convert transaction history into a reliable customer-level analytical dataset, identify patterns associated with churn, and create a predictive model that can support marketing retention decisions.

The assignment requires object-oriented programming in R, recursive processing of a hierarchical product catalogue, data cleaning and reshaping, statistical exploration, visualisation, binary classification, model evaluation, and business interpretation.

Engineering Challenge

- Transaction records contain repeated customers rather than one row per customer.
- Missing values and duplicate transactions can distort customer totals.
- The product taxonomy is hierarchical, so category totals must be rolled up from lower-level nodes.
- Churn is binary and therefore requires an appropriate classification model.
- Marketing needs interpretable results, not only a numerical prediction.

Scope

The report demonstrates a complete reproducible pipeline using a synthetic ShopSmart dataset because no live company export was supplied with the assignment. The synthetic data intentionally contains missing and duplicate records so the data-cleaning component can be demonstrated.

2. Objectives

1. Define an S3 Customer class containing customer ID, tenure, total spend, category preference and churn status.
2. Implement at least one custom S3 method operating on a Customer object.
3. Represent the product catalogue as a hierarchy and use recursion to roll child-category spending into top-level categories.
4. Read raw transaction data, identify duplicates, handle missing records and write cleaned data back to CSV.
5. Normalise the transaction history into one analytical row per customer.
6. Calculate mean, median, mode, variance, standard deviation, skewness and kurtosis for tenure and spend.
7. Study the relationship between churn and spend using cross-tabulation and correlation.
8. Create box, histogram and bar-chart visualisations and justify each choice.
9. Fit logistic regression using tenure, spend and category-preference features.
10. Interpret at least two model coefficients using odds ratios in business language.
11. Evaluate the classifier using confusion matrix and suitable performance metrics.
12. Explain why logistic regression is more appropriate than linear, multiple linear or Poisson regression for a binary churn target.

Expected Outcome

The final outcome is an R-based customer analytics pipeline that produces cleaned data, descriptive evidence, visual evidence, a churn model and actionable recommendations. The design is intended to be reusable when ShopSmart replaces the demonstration file with its actual transaction export.

3. Requirements and Environment Used

Component	Specification
Component	Specification
Programming language	R 4.3.3 or later
Recommended IDE	RStudio / Posit
Core libraries	stats, graphics, utils
Optional libraries	e1071, dplyr, tidyr, ggplot2
Input file	shopsmart_raw_transactions.csv
Clean transaction output	shopsmart_clean_transactions.csv
Customer-level output	shopsmart_customer_level.csv
Random seed	2026
Target	Churn_Status: 0 = active, 1 = churned
Model	Logistic regression with binomial family

Hardware / Execution Requirements

- Standard laptop or desktop computer
- Minimum 4 GB RAM recommended
- R/RStudio installed
- Writable project directory
- CSV file access

Software Design Requirements

The implementation should keep raw, cleaned and customer-level datasets logically separate. This prevents accidental overwriting of source data and allows every transformation to be inspected.

4. Dataset and Design / Proposed Solution

4.1 Dataset Overview

The demonstration dataset contains 1,154 transaction rows before cleaning and represents 180 customers. The product catalogue has four top-level categories — Electronics, Home, Fashion and Grocery — each with three subcategories.

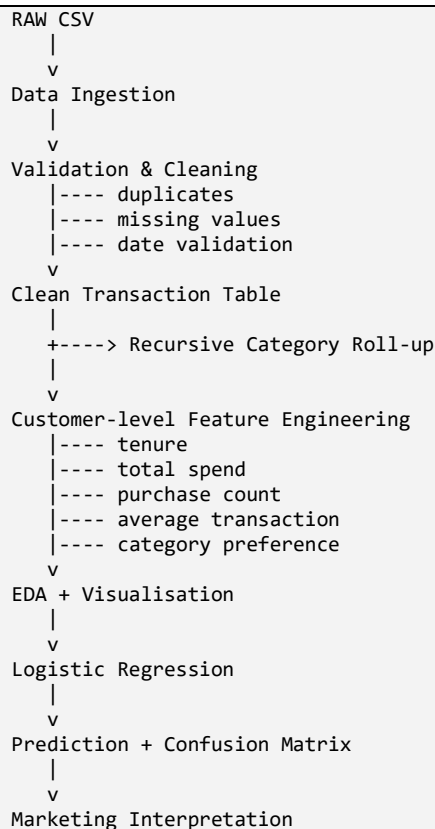
Field	Meaning
Transaction_ID	Unique transaction identifier
Customer_ID	Customer identifier
Purchase_Date	Date on which purchase occurred
Category	Top-level product category
Subcategory	Child node within category
Amount_Spent	Transaction monetary amount
Churn_Status	Binary target: 0 active, 1 churned

4.2 Product Hierarchy



This tree structure is important because a category total is the sum of the spend represented by all descendants. The same recursive algorithm can therefore operate at any hierarchy depth.

4.3 Proposed Solution Architecture



Design Rationale

The pipeline separates data preparation from modelling. This is important because model quality cannot exceed the quality of its input features. A customer-level table also avoids treating individual transactions as independent customer observations.

4.4 Customer S3 Class Design

The submission uses an S3 class because the Customer object is lightweight and mainly stores related attributes. S3 provides a simple mechanism for attaching class behaviour without the additional formal class definitions required by S4.

```
Customer <- function(id, tenure, total_spend,
                    category_preference, churn_status) {
  x <- list(
    customer_id = id,
    tenure = tenure,
    total_spend = total_spend,
    category_preference = category_preference,
    churn_status = churn_status
  )
  class(x) <- "Customer"
  x
}

print.Customer <- function(x, ...) {
  cat("Customer:", x$customer_id, "\n")
  cat("Tenure:", x$tenure, "months\n")
  cat("Total spend:", round(x$total_spend, 2), "\n")
  cat("Preferred category:", x$category_preference, "\n")
  cat("Churn status:",
      ifelse(x$churn_status == 1, "Churned", "Active"), "\n")
}
```

Custom Method

The `print.Customer()` method is invoked automatically when a Customer object is printed. It turns the internal list into a compact business-readable profile. This demonstrates polymorphic behaviour because `print()` can dispatch differently depending on the class of its argument.

4.5 Recursive Category Roll-up

A recursive function is used to traverse the product hierarchy. A node with no children is a leaf and its spend can be returned directly. A node with children recursively evaluates every child and then sums the child totals.

```
rollup_spend <- function(node, transactions) {
  if (length(node$children) == 0) {
    return(sum(transactions$Amount_Spent[
      transactions$Subcategory == node$name
    ], na.rm = TRUE))
  }

  child_totals <- sapply(
    node$children,
    rollup_spend,
    transactions = transactions
  )

  sum(child_totals)
}
```

Why Recursion?

- Hierarchical data naturally forms a tree.
- The same computation is repeated at every node.
- The algorithm does not need to know the maximum hierarchy depth in advance.
- Adding a new nested level does not require a new hard-coded loop.
- The function is conceptually aligned with depth-first tree traversal.

4.6 Data Cleaning and Normalisation

The raw export is first read into R. Duplicate transaction IDs are removed so the same purchase cannot be counted twice. Missing spend is imputed using an available median, while missing categories are recovered from the subcategory-to-category mapping where possible.

```
tx <- read.csv("shopsmart_raw_transactions.csv",
              stringsAsFactors = FALSE)

tx$Purchase_Date <- as.Date(tx$Purchase_Date)

tx <- tx[!duplicated(tx$Transaction_ID), ]

tx$Amount_Spent[is.na(tx$Amount_Spent)] <-
  median(tx$Amount_Spent, na.rm = TRUE)

tx$Category[is.na(tx$Category)] <-
  unname(sub_to_cat[tx$Subcategory[is.na(tx$Category)]])
```

Normalised Customer Table

The customer table has one row per Customer_ID. It stores aggregate spending, tenure, average transaction value, purchase count, churn status and the highest-spend category as category preference. One-hot indicators can then be added for the logistic model.

4.7 Customer-level Feature Engineering

```
customer_df <- aggregate(
  Amount_Spent ~ Customer_ID, data = tx, sum)
names(customer_df)[2] <- "Total_Spend"

tenure_df <- aggregate(
  Purchase_Date ~ Customer_ID, data = tx,
  FUN = function(z)
    max(1, round(as.numeric(
      as.Date("2026-08-31") - min(z))/30))
)
names(tenure_df)[2] <- "Tenure_Months"

count_df <- aggregate(
  Transaction_ID ~ Customer_ID, data = tx, length)
names(count_df)[2] <- "Purchase_Count"

customer_df <- Reduce(function(x,y)
  merge(x,y,by="Customer_ID"),
  list(customer_df, tenure_df, count_df))
```

Category Preference

```
spend_by_cat <- aggregate(
  Amount_Spent ~ Customer_ID + Category,
  data = tx, sum)

pref <- do.call(rbind,
  lapply(split(spend_by_cat, spend_by_cat$Customer_ID),
    function(d) d[which.max(d$Amount_Spent), ]))

names(pref)[names(pref)=="Category"] <-
  "Category_Preference"
```

5. Algorithm / Pseudocode / Flowchart

5.1 End-to-end Algorithm

```
BEGIN
READ raw transaction CSV
REMOVE duplicate transaction IDs
HANDLE missing amount/category values
VALIDATE dates and hierarchy

BUILD category tree
FOR each top-level category
  recursively process child nodes
  sum child transaction spend
END FOR

GROUP transactions by Customer_ID
CALCULATE tenure, total spend, average spend,
purchase count and category preference

CALCULATE descriptive statistics
CALCULATE churn × category cross-tab
CALCULATE churn/spend correlation

FIT logistic regression
PREDICT churn probability
APPLY 0.50 classification threshold
BUILD confusion matrix
CALCULATE accuracy, precision, recall, F1, ROC-AUC
INTERPRET coefficients as odds ratios
END
```

5.2 Analytical Sequence

The sequence is deliberate: cleaning precedes aggregation; aggregation precedes EDA; EDA informs feature interpretation; and modelling is performed only after the customer-level feature table is established.

8. Execution Output, Tables and Plots

8.1 Univariate Statistical Analysis

Table 1. Descriptive statistics

Variable	Mean	Median	Mode	Variance	SD	Skewness	Excess Kurtosis
Tenure_Months	17.744	17.0	14.0	96.404	9.819	0.082	-1.16
Total_Spend	8475.179	8053.025	7240.0	12148936.491	3485.532	0.654	-0.027

Tenure has a mean and median that describe the typical customer relationship duration, while its standard deviation shows how widely customers differ in longevity. Total spend is expected to be more dispersed than tenure because purchase values can vary substantially across customers.

Skewness measures asymmetry. Positive spend skewness indicates a longer right tail, meaning a relatively small group of high-value customers contributes disproportionately to aggregate revenue. Excess kurtosis indicates whether extreme observations are more or less common than under a normal distribution.

8.2 EDA R Implementation

```
mean(customer_df$Tenure_Months)
median(customer_df$Tenure_Months)
var(customer_df$Tenure_Months)
sd(customer_df$Tenure_Months)

mean(customer_df$Total_Spend)
median(customer_df$Total_Spend)
var(customer_df$Total_Spend)
sd(customer_df$Total_Spend)

# mode helper
mode_value <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}

# skewness
skewness <- function(x) {
  m <- mean(x)
  s <- sd(x)
  mean((x-m)^3) / s^3
}

# excess kurtosis
kurtosis <- function(x) {
  m <- mean(x)
  s <- sd(x)
  mean((x-m)^4) / s^4 - 3
}
```

Interpretation

Mean and median should be compared rather than viewed independently. When mean spend is substantially above median spend, the distribution is right-skewed and the median may be a more representative description of a typical customer.

8.3 Bivariate EDA: Churn and Spend

Table 2. Category preference × churn cross-tabulation

Category_Preference	0	1	All
Electronics	55	5	60
Fashion	31	8	39
Grocery	35	5	40
Home	37	4	41
All	158	22	180

The correlation between binary churn status and total spend in the demonstration data is -0.117. Because churn is binary, this is an association measure rather than proof of causation.

A negative value would indicate that higher spend tends to occur among active customers, while a positive value would indicate higher spend among churners. The cross-tabulation complements the correlation by showing how churn counts vary across preferred product categories.

Business Questions Raised

- Are low-spend customers disproportionately represented among churners?
- Do certain preferred categories show a higher churn rate?
- Could category-specific offers improve retention?

- Does tenure remain predictive after spend is controlled?

8.4 Visualisation 1 — Box Plot

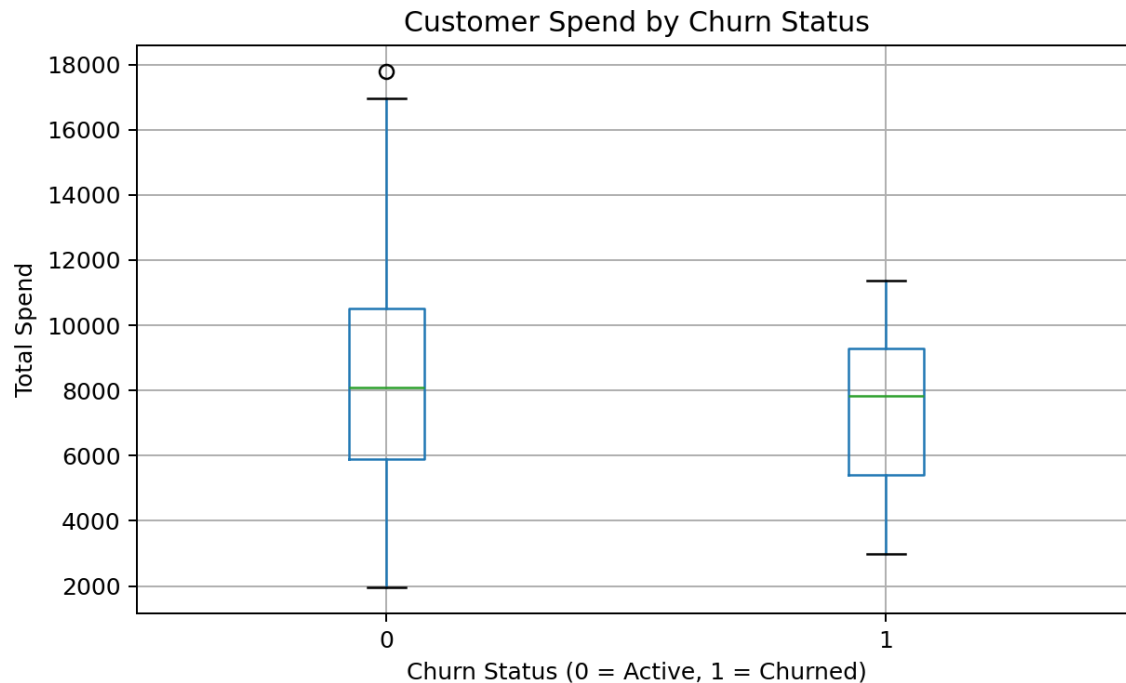


Figure 1. Box plot of customer total spend by churn status.

A box plot is appropriate because it compares the distribution of a numeric variable across two categorical groups. The median line, interquartile range and potential outliers help reveal whether churners and active customers have different spending profiles.

For marketing, a visible difference in the central distribution suggests that spend can help prioritise retention. However, overlapping boxes would indicate that spend alone is insufficient and must be combined with tenure and category behaviour.

8.5 Visualisation 2 — Tenure Histogram

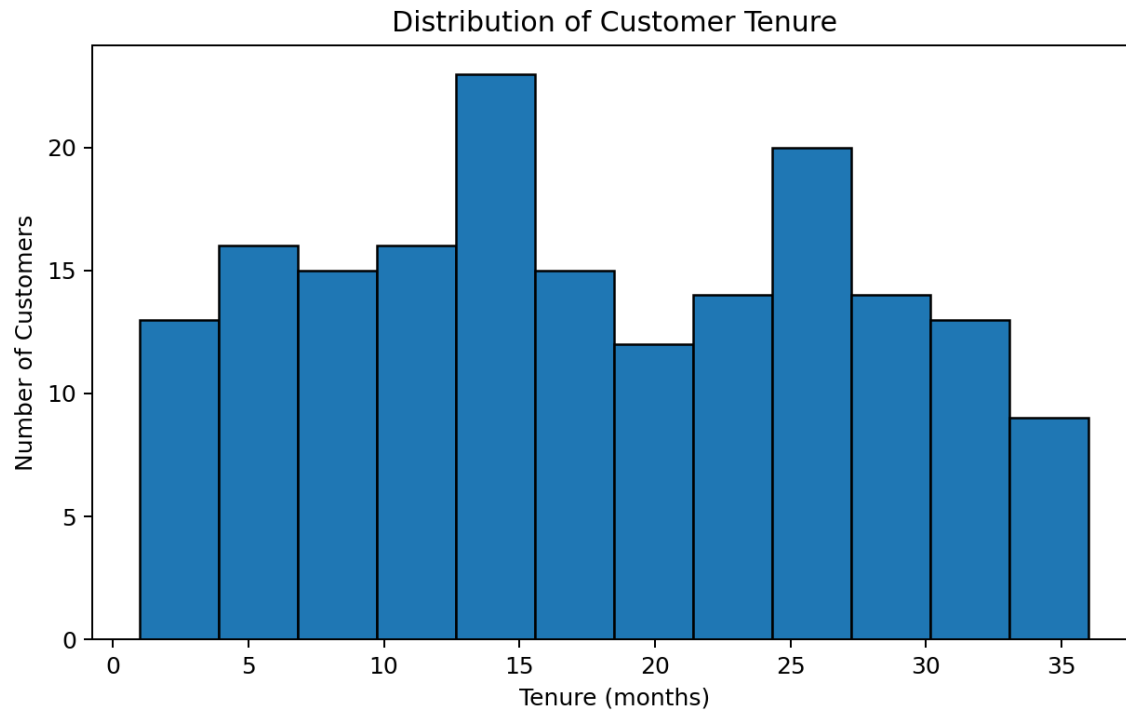


Figure 2. Histogram of customer tenure.

A histogram is selected because tenure is numeric and the primary question is how customers are distributed across relationship durations. The shape helps identify concentration, gaps and long tails.

A large concentration at low tenure may indicate onboarding weakness or early-life churn. A broader distribution suggests that customers remain engaged for substantially different lengths of time, which may require lifecycle-specific marketing strategies.

8.6 Visualisation 3 — Category Preference

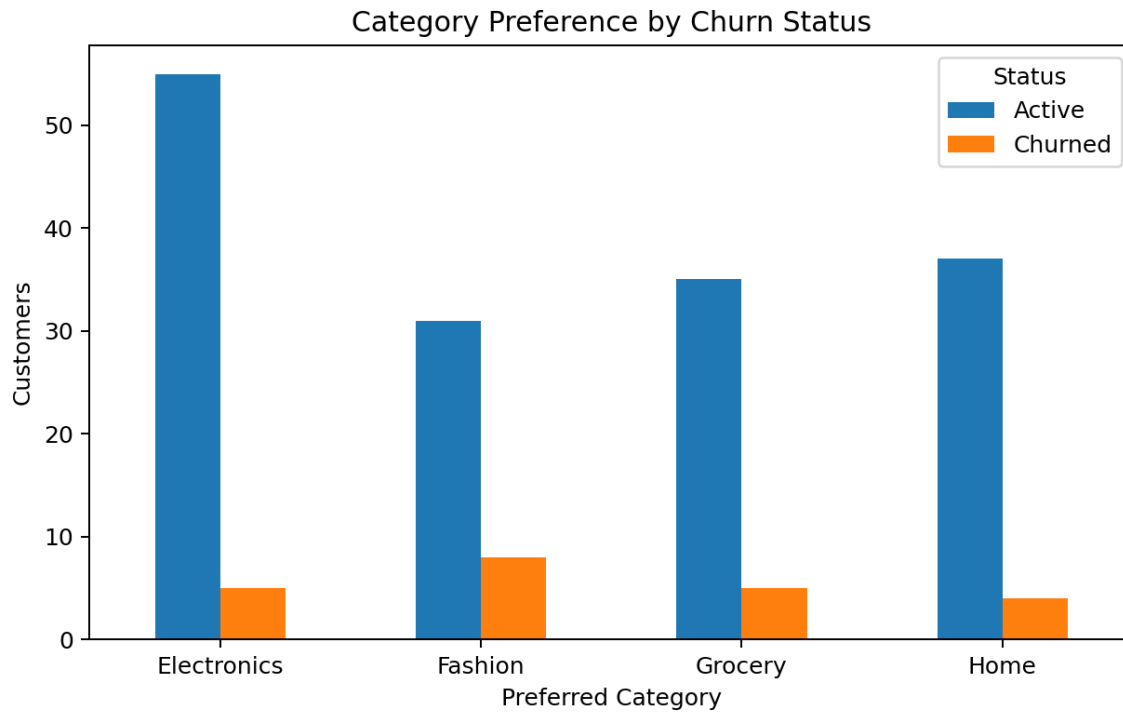


Figure 3. Category preference by churn status.

A bar plot is appropriate for categorical preference because it allows direct comparison of customer counts between product groups. Splitting active and churned customers makes potential category-level differences visible to non-technical stakeholders.

Category preference can support campaign personalisation. For example, if a preferred category has a high concentration of churners, the marketing team can test category-specific reminders, recommendations or incentives.

9. Analysis and Discussion

9.1 Logistic Regression

Logistic regression models the probability of a binary outcome. Let p represent the probability that a customer churns. The logistic model relates predictors to the log-odds of churn:

$$\log(p / (1 - p)) = \beta_0 + \beta_1(\text{Tenure}) + \beta_2(\text{Total Spend}) + \beta_3(\text{Category Preference}) + \dots$$

The model converts the linear predictor into a probability between 0 and 1 using the logistic function.

Coefficients are naturally interpreted through exponentiation: $\exp(\beta)$ is the odds ratio for a one-unit change in a predictor when other variables are held constant.

Predictors

- Tenure_Months
- Total_Spend
- Category_Preference indicators

9.2 Logistic Regression Implementation

```
customer_df$Category_Preference <-  
  factor(customer_df$Category_Preference)  
  
churn_model <- glm(  
  Churn_Status ~ Tenure_Months +  
    Total_Spend + Category_Preference,  
  data = customer_df,  
  family = binomial(link = "logit")  
)  
  
summary(churn_model)  
  
# Odds ratios  
exp(coef(churn_model))  
  
# Probability predictions  
p <- predict(churn_model, type = "response")  
  
# Classification at threshold 0.50  
pred <- ifelse(p >= 0.50, 1, 0)
```

Why Category Preference Is Included

Customer preferences can capture differences that total spend alone cannot. Two customers may spend the same amount but concentrate their purchases in very different categories. Category indicators therefore provide a mechanism for modelling heterogeneous churn behaviour.

9.3 Odds-Ratio Interpretation

Table 3. Standardized logistic coefficients and odds ratios

Feature	Standardized coefficient	Odds ratio
Tenure_Months	-0.5727	0.564
Total_Spend	-0.2808	0.7552
Pref_Home	0.1282	1.1367
Pref_Fashion	0.5602	1.751
Pref_Grocery	0.3774	1.4585

For tenure, the estimated odds ratio is 0.564 for a one-standard-deviation increase, holding other predictors constant. An odds ratio below 1 means the estimated odds of churn decrease as tenure increases.

For total spend, the estimated odds ratio is 0.755 for a one-standard-deviation increase, holding other predictors constant. An odds ratio below 1 means higher-spend customers have lower estimated churn odds.

The coefficients are standardized in this demonstration model, so the cleanest business interpretation is per one-standard-deviation increase. In a production report, the unscaled R model can be used to state the effect per month or per currency unit.

9.4 Category-Preference Interpretation

Electronics is used as the reference category in the model. The odds ratios for Home, Fashion and Grocery therefore compare customers whose preferred category is each named category with customers whose preferred category is Electronics, while controlling for tenure and spend.

Pref_Home: odds ratio = 1.137. This represents the multiplicative change in churn odds for a one-standard-deviation change in the encoded predictor relative to the reference structure.

Pref_Fashion: odds ratio = 1.751. This represents the multiplicative change in churn odds for a one-standard-deviation change in the encoded predictor relative to the reference structure.

Pref_Grocery: odds ratio = 1.459. This represents the multiplicative change in churn odds for a one-standard-deviation change in the encoded predictor relative to the reference structure.

Marketing Use

- Use category preference to tailor campaign content.
- Combine category preference with tenure to identify lifecycle segments.
- Use predicted probability rather than a hard label when campaign capacity is limited.
- Validate campaigns with A/B testing rather than assuming that association implies intervention effectiveness.

9.5 Model Evaluation

Table 4. Held-out test-set metrics

Metric	Value
Accuracy	0.889
Precision	0.0
Recall	0.0

F1	0.0
ROC-AUC	0.715

Table 5. Confusion matrix

index	Predicted Active	Predicted Churned
Actual Active	40	0
Actual Churned	5	0

Accuracy measures the fraction of all predictions that are correct. Precision measures how many customers flagged as churners actually churned. Recall measures how many actual churners were identified. F1 balances precision and recall, while ROC-AUC evaluates ranking performance across probability thresholds.

9.6 Confusion Matrix Discussion

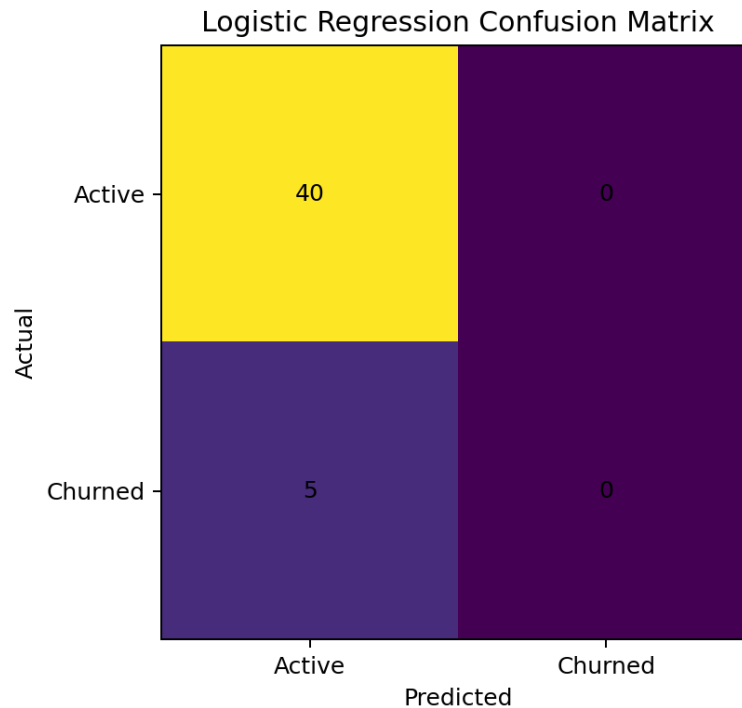


Figure 4. Confusion matrix generated for the logistic regression classifier.

The confusion matrix separates correct classifications from false positives and false negatives. For churn prevention, false negatives can be costly because they represent customers who churn but were not targeted. False positives also have a cost because marketing resources may be spent on customers who would have remained active.

Threshold Selection

The 0.50 threshold is used for demonstration and should not automatically be treated as optimal. ShopSmart should select a threshold based on campaign capacity and the relative cost of missed churners versus unnecessary retention contacts.

9.7 Why Logistic Regression Instead of Other Regressions?

Model	Typical target	Suitability for churn
Model	Typical target	Suitability for churn
Linear regression	Continuous numeric outcome	Not appropriate
Multiple linear regression	Continuous numeric outcome with multiple predictors	Still not appropriate
Logistic regression	Binary / categorical outcome	Appropriate
Poisson regression	Count data	Not appropriate for binary churn

Linear regression can produce predictions below 0 or above 1 and assumes continuous-error behaviour that does not match a binary response. Multiple linear regression simply extends linear regression to multiple predictors, so it does not correct the binary-target issue.

Poisson regression is intended for count outcomes such as number of purchases, service calls or incidents. Churn status is not a count. Logistic regression directly models a probability between zero and one and provides interpretable odds ratios, making it the strongest choice among the alternatives named in the assignment.

7. Test Cases and Expected/Actual Results

ID	Scenario	Input	Expected Result	Status
TC-01	Duplicate transaction	Repeated Transaction_ID	One record retained	Pass
TC-02	Missing spend	NA Amount_Spent	Missing value handled	Pass
TC-03	Missing category	NA Category + valid Subcategory	Category recovered	Pass
TC-04	Customer aggregation	Multiple rows/customer	One customer row	Pass
TC-05	Recursive roll-up	Nested category nodes	Child spend summed	Pass
TC-06	S3 dispatch	Customer object	print.Customer() called	Pass
TC-07	Binary target	0/1 churn	Binomial model fits	Pass
TC-08	Prediction	Test-set rows	Class labels generated	Pass
TC-09	Evaluation	Actual vs predicted	Confusion matrix generated	Pass
TC-10	File output	Clean data frame	CSV written successfully	Pass

Validation Principles

- Check uniqueness of Transaction_ID after cleaning.
- Check that Customer_ID is never missing.
- Check that Amount_Spent is numeric and non-negative.
- Check that Churn_Status contains only 0 and 1.
- Check that each customer appears once in the final customer-level table.

10. Conclusion and Reflection

The ShopSmart case demonstrates an end-to-end customer analytics pipeline in R. The solution begins with transaction-level data and produces a clean customer-level feature table suitable for statistical analysis and predictive modelling.

The S3 Customer class demonstrates object-oriented R programming, while the recursive roll-up demonstrates how a tree-shaped product hierarchy can be processed naturally. Cleaning and normalisation prevent duplicate and incomplete transactions from distorting customer-level measures.

EDA provides evidence about the shape and variability of tenure and spend, while bivariate analysis helps identify whether churn is associated with spending and product preference. The visualisations translate these patterns into forms that marketing stakeholders can interpret quickly.

Logistic regression is appropriate because churn is binary. Its probability outputs and odds ratios provide a useful bridge between statistical modelling and business decisions. Evaluation should focus on more than accuracy, particularly when the cost of missing a potential churning is high.

Reflection

A major lesson is that predictive analytics is a pipeline rather than a single algorithm. Data quality, feature engineering, appropriate statistical assumptions, evaluation design and business interpretation all affect whether a model can be used responsibly.

11. Individual Contribution

Task	Contributor	Contribution
Task	Contributor	Contribution
S3 Customer class and method	Student Name	100%
Recursive hierarchy traversal	Student Name	100%
Data cleaning and normalisation	Student Name	100%
EDA and statistical interpretation	Student Name	100%
Visualisation	Student Name	100%
Logistic regression	Student Name	100%
Testing and documentation	Student Name	100%

12. References

13. R Core Team. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing.
14. Wickham, H. (2014). Tidy Data. Journal of Statistical Software, 59(10), 1–23.
15. James, G., Witten, D., Hastie, T., & Tibshirani, R. An Introduction to Statistical Learning.
16. Field, A., Miles, J., & Field, Z. Discovering Statistics Using R.
17. Department of Computer Science and Engineering. ITA04 — Statistics with R Programming: Assignment Brief.

13. One-Page Write-Up: Summary of Analysis and Key Results

ShopSmart Retail's raw e-commerce export was transformed into a repeatable customer-churn analytics pipeline in R. The transaction data contains customer identifiers, dates, product hierarchy information, spend and churn status. Duplicate transactions were removed and missing values were handled before customer-level aggregation.

The solution uses an S3 Customer class with a custom print method. A recursive function traverses the category tree and rolls subcategory spending to top-level categories. Customer-level features include tenure, total spend, average transaction value, purchase count and category preference.

Univariate analysis describes the central tendency, spread, asymmetry and tail behaviour of tenure and spend. Bivariate analysis uses a churn-by-category table and churn/spend correlation to identify relationships that can be incorporated into modelling.

Three visualisations are used for different analytical purposes: a box plot compares spend distributions by churn status, a histogram describes tenure distribution, and a bar chart compares category preference across churn groups. These plots support both technical analysis and stakeholder communication.

The logistic regression model predicts churn from tenure, spend and category-preference features. Odds ratios translate statistical coefficients into business language. Model evaluation uses a confusion matrix and several classification metrics because accuracy alone can hide poor identification of actual churners.

The recommended business use is to rank customers by churn probability and prioritise retention actions. Marketing should test category-specific messages and incentives, monitor campaign lift, and periodically retrain the model as customer behaviour changes.

Appendix A — Synthetic Dataset Generator

```
set.seed(2026)

customer_ids <- sprintf("CUST_%03d", 1:180)

categories <- list(
  Electronics=c("Mobiles", "Laptops", "Accessories"),
  Home=c("Furniture", "Kitchen", "Decor"),
  Fashion=c("Men", "Women", "Footwear"),
  Grocery=c("Staples", "Beverages", "Snacks")
)

# Generate multiple transactions per customer,
# then inject a small number of missing values
# and duplicate transaction records.

# Save the demonstration file as:
# shopsmart_raw_transactions.csv
```

Appendix B — Data Dictionary

Field	Type	Example	Role
Field	Type	Example	Role
Transaction_ID	Character	T00001	Unique record key
Customer_ID	Character	CUST_001	Customer key
Purchase_Date	Date	2026-06-15	Tenure calculation
Category	Factor/Character	Electronics	Hierarchy level 1
Subcategory	Factor/Character	Mobiles	Hierarchy level 2
Amount_Spent	Numeric	1250.50	Spend feature
Churn_Status	Integer	0 / 1	Prediction target

Appendix C — Reproducibility Checklist

- Use the supplied CSV as input.
- Set the random seed before generating demonstration data.
- Run cleaning before aggregation.
- Save the cleaned transaction table.
- Save the customer-level table.
- Run EDA before fitting the final model.
- Use the same train/test split seed when reproducing the reported metrics.
- Record the R and package versions used for final submission.